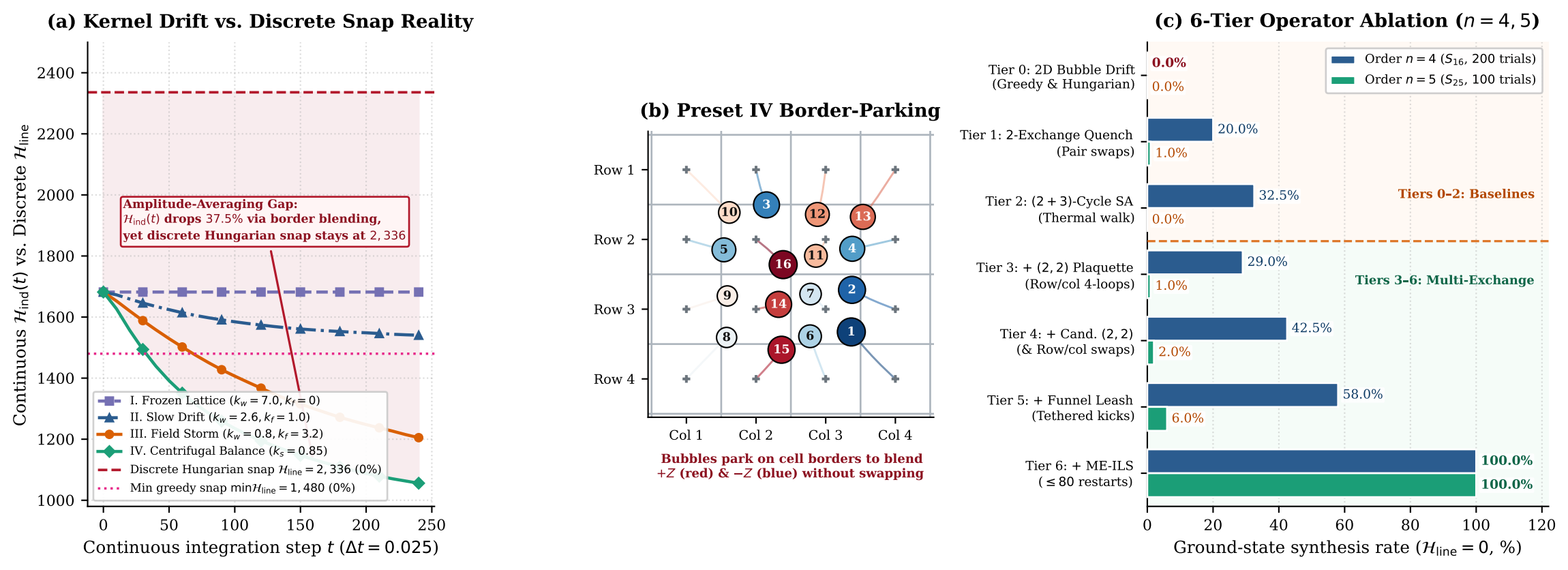

(a) Kernel Drift vs. Discrete Snap Reality
Continuous $\mathcal{H}_{\text{ind}}(t)$ vs. Discrete $\mathcal{H}_{\text{line}}$
Amplitude-Averaging Gap:
$\mathcal{H}_{\text{ind}}(t)$ drops 37.5% via border blending,
yet discrete Hungarian snap stays at 2,336
I. Frozen Lattice ($k_w = 7.0, k_f = 0$)
II. Slow Drift ($k_w = 2.6, k_f = 1.0$)
III. Field Storm ($k_w = 0.8, k_f = 3.2$)
IV. Centrifugal Balance ($k_s = 0.85$)
Discrete Hungarian snap $\mathcal{H}_{\text{line}} = 2,336$ (0%)
Min greedy snap $\min\mathcal{H}_{\text{line}} = 1,480$ (0%)
Continuous integration step t ($\Delta t = 0.025$)
(b) Preset IV Border-Parking
Row 1
Row 2
Row 3
Row 4
Col 1
Col 2
Col 3
Col 4
Bubbles park on cell borders to blend
$+Z$ (red) & $-Z$ (blue) without swapping
(c) 6-Tier Operator Ablation ($n = 4, 5$)
Order $n = 4$ (S_{16}, 200 trials)
Order $n = 5$ (S_{25}, 100 trials)
Tier 0: 2D Bubble Drift
(Greedy & Hungarian)
Tier 1: 2-Exchange Quench
(Pair swaps)
Tier 2: (2 + 3)-Cycle SA
(Thermal walk)
Tier 3: + (2, 2) Plaquette
(Row/col 4-loops)
Tier 4: + Cand. (2, 2)
(& Row/col swaps)
Tier 5: + Funnel Leash
(Tethered kicks)
Tier 6: + ME-ILS
(≤ 80 restarts)
0.0%
0.0%
20.0%
1.0%
32.5%
0.0%
29.0%
1.0%
42.5%
2.0%
58.0%
6.0%
100.0%
100.0%
Tiers 0–2: Baselines
Tiers 3–6: Multi-Exchange
Ground-state synthesis rate ($\mathcal{H}_{\text{line}} = 0$, %)